MEDIUM

ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

03

10 questions · ~15 min

SAT QUESTION BANK

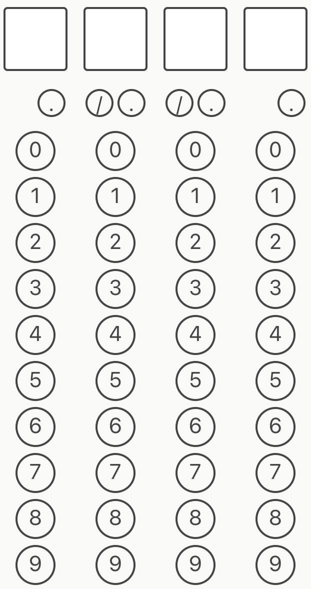
2026-05-29

Q1 GRID-IN ADVANCED MATH

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ADVANCED MATH

$$\frac{x + 9x - 9}{x + 9} = 7$$

What is the solution to the given equation?

A. 7

B. 9

C. 16

D. 63

Q3 ADVANCED MATH

The speed of sound in dry air, v, can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

A. $T = \dfrac{v + 0.606}{331.3}$

B. $T = \dfrac{v - 0.606}{331.3}$

C. $T = \dfrac{v + 331.3}{0.606}$

D. $T = \dfrac{v - 331.3}{0.606}$

Q4 ADVANCED MATH

An oceanographer uses the equation $s=\frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

A. $p=\frac{2}{3}s$

B. $p=\frac{3}{2}s$

C. $p=\frac{2}{3}+s$

D. $p=\frac{3}{2}+s$

Q5 ADVANCED MATH

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A. $x = \frac{b - 72}{y}$

B. $x = by - 72$

C. $x = \frac{by - 72}{y}$

D. $x = by - 72y$

Q6 ADVANCED MATH

$$x^2 = -841$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q7

GRID-IN ADVANCED MATH

In the xy-plane, what is the y-coordinate of the point of intersection of the graphs of $y=(x-1)^2$ and $y=2x-3$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN ADVANCED MATH

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

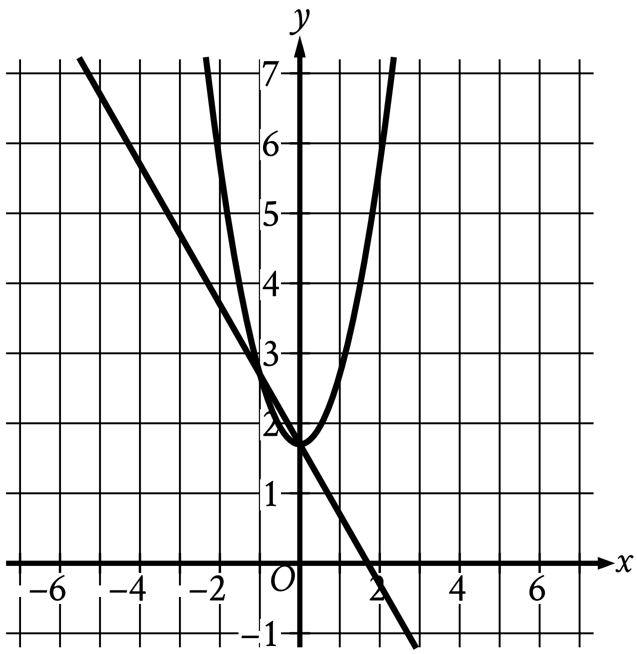
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

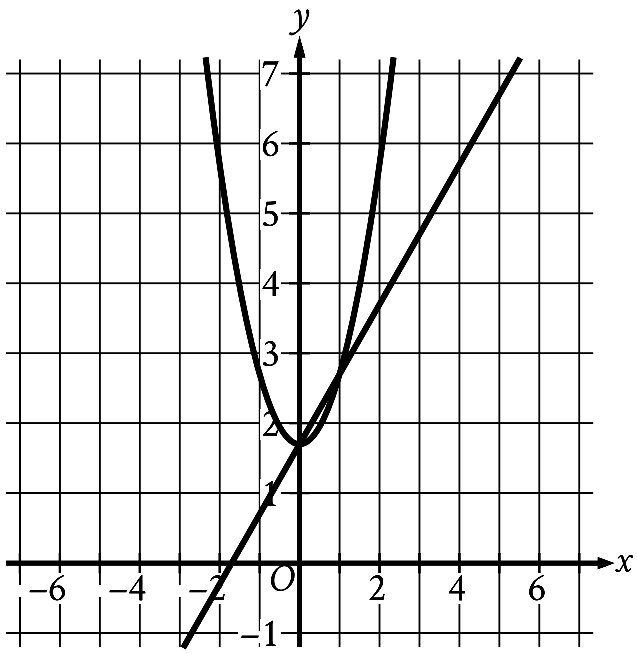
Which graph represents the given system of equations?

A.



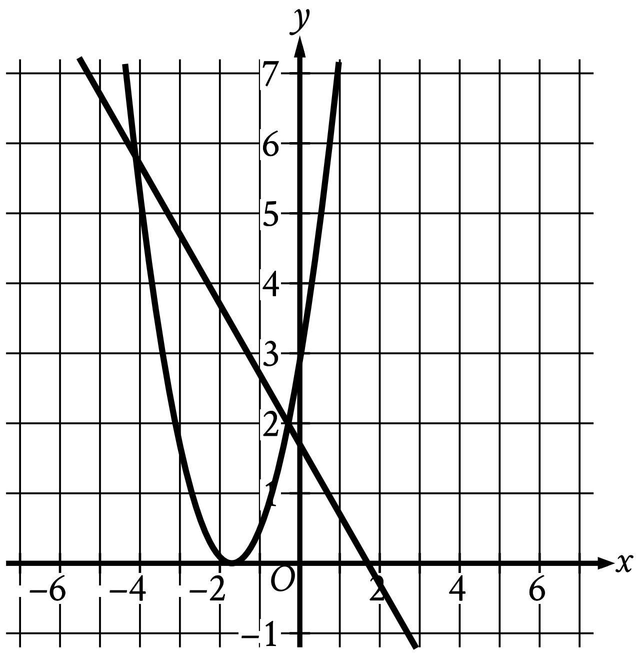
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

B.



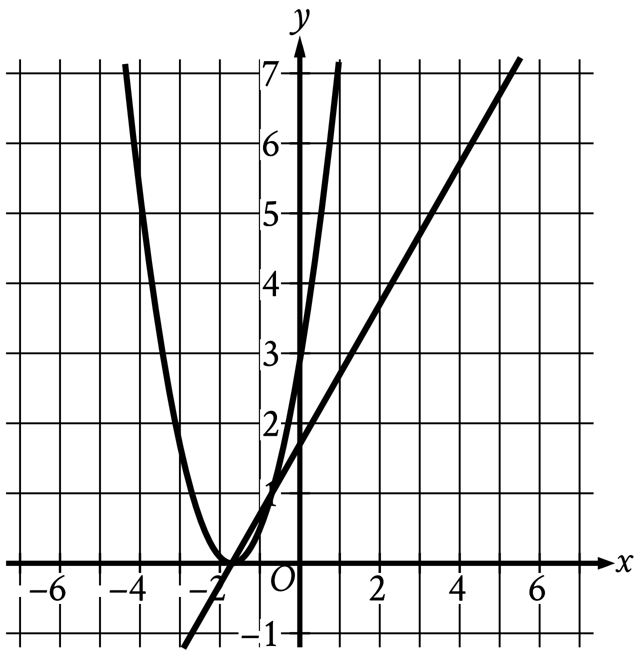
- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

C.



- For the line in the system:
    - The line slants sharply down from left to right.
    - The line passes through the following points:
        - (negative 1 comma 2.7)
        - (0 comma 1.7)
- For the parabola in the system:
    - The parabola opens upward.
    - The vertex is at point (negative 1.7 comma 0).
    - The parabola passes through the following points:
        - (negative 2.7 comma 1)
        - (negative 1.7 comma 0)
        - (negative 0.7 comma 1)

D.



- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)

Q10 ADVANCED MATH

$$x^2 - x - 1 = 0$$

What values satisfy the equation above?

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$